PENGWIN 2026 Task 1 -- V0.2 Pipeline-Test Submission

Algorithm Description · 2026-05-30

0. Submission intent

This is a **V0.2 pipeline-test** submission for the PENGWIN 2026 Task 1 preliminary phase. The objective is to validate the end-to-end Grand Challenge contract (Docker container build, model tarball, `/input` MHA read, `/output` MHA write under `--network none`) with the current best in-house two-stage pipeline. It is **not** a competitive scientific submission and should not be interpreted as our final leaderboard entry. The femur anatomy (label range 151-200) is **intentionally not modeled** in V0.2; femur fragments are emitted as background (0). Femur modeling is planned for V1.

1. Method overview

V0.2 is a **two-stage anatomy-conditioned pipeline**. Stage 1 is a whole-CT anatomy classifier that produces a 4-class softmax (background / Sacrum / LeftHipbone / RightHipbone). Stage 2 is a per-anatomy ABBC nnU-Net that takes a **3-channel ROI** -- the bone-LUT CT crop, the Stage 1 anatomy probability crop, and a signed distance map derived from that probability -- and produces a 4-class ABBC softmax (background / border / boundary / core). Per-instance fragments are recovered by a core-seed watershed decoder and pasted back into the full label volume.

Components:

- **Stage 1 -- anatomy classifier:** nnU-Net ResEnc-L trained on `Dataset532_PelvicAnatomyV2`. Input is the full CT in LPS orientation, normalized with the bone HU window ([-1000, 2000] clip + bone-LUT). Output is a 4-class softmax over background / Sacrum / LeftHipbone / RightHipbone.
- **Per-anatomy ROI extraction:** for each of {Sacrum, LeftHipbone, RightHipbone}, take the bounding box of the Stage 1 `prob >= 0.5` mask, pad by 24 voxels on each axis, and build a **3-channel ROI**: 1. bone-LUT normalized CT crop, 2. Stage 1 anatomy probability crop (the channel for this anatomy), 3. signed distance to the `prob >= 0.5` surface, clipped to +/- 40 mm.
- **Stage 2 -- ABBC fragment model:** V291 ABBC nnU-Net ResEnc-L (`nnUNetResEncUNetLPlans`, `3d_fullres`), 3-channel input, 4-class ABBC softmax (background / border / boundary / core).
- **Instance decoder:** V288-style core-seed watershed lifts the 4-class softmax to per-instance integer fragments, followed by ~1 cm^3 connected-component prune and an anatomy-size-ratio merge (`sr_keep = 0.05`).
- **Label assembly:** per-anatomy fragments are remapped into the PENGWIN label ranges (Sacrum 1-50, LeftHipbone 51-100, RightHipbone 101-150) and pasted into the full-volume label map. The label map is then reoriented back from LPS to the original input frame.

Femur (151-200) is **intentionally unmodeled** in V0.2 and is written as background; a femur stage is planned for V1.

1.5 V0.3 robustness layers

V0.3.1 wraps the V0.2 two-stage pipeline in a **4-layer robustness stack** designed to keep the container alive and on-budget across out-of-distribution Grand Challenge inputs. V0.2 had a known GC failure mode: on small-FOV inputs the Ds532 Stage 1 anatomy classifier fired out-of-distribution and produced large spurious anatomy masks, which caused the per-anatomy ROI to balloon, blow past the 10-minute GC time budget, and time out the case. V0.3.1 fixes this by adding distribution-independent fallbacks, mutually-exclusive anatomy assignment, post-pad bbox sanity, and an explicit time budget.

- **L1 -- bone-skeleton anatomy decomposition** (HU>200 connected components). A distribution-independent fallback derived directly from the CT (bone-threshold + 3D connected components) that produces a coarse Sacrum / LeftHipbone / RightHipbone partition without depending on the Ds532 classifier. L1 is always available as a fallback when L2 misfires or is suppressed by L3.
- **L2 -- Ds532 argmax masks** (mutually-exclusive anatomy assignment). Replaces the V0.2 per-channel `prob >= 0.5` threshold (which could double-assign voxels to overlapping anatomies and produce inflated ROIs) with a single `argmax` over the 4-class Stage 1 softmax, yielding mutually-exclusive Sacrum / LeftHipbone / RightHipbone masks.

- **L3 -- bbox-fraction sanity** (post-pad bbox >50% triggers pad-shrink: 24 -> 12 -> 6 -> 0 vox). After padding the L2 argmax bounding box by 24 voxels, if the resulting box covers more than 50% of the full CT volume the pad is shrunk through the cascade 24 -> 12 -> 6 -> 0 voxels until the post-pad bbox falls under the 50% threshold. This catches Ds532 OOD misfires before they can enter Stage 2.
- **L4 -- time-budget enforcement** (8 min soft, 9 min hard). A wall-clock budget tracked across the per-anatomy loop: at 8 minutes the pipeline switches to a soft-degraded path (skips remaining optional refinement), at 9 minutes it hard-stops and emits the best label map assembled so far rather than timing out the GC job at 10 minutes.

V0.3.1 directly addresses V0.2's GC failure mode -- Ds532 OOD misfire on small-FOV inputs causing a 10-minute timeout -- through the L2 argmax fix (no double-assignment), the L3 bbox-fraction sanity check (shrinks the pad before the ROI blows up), the L1 bone-skeleton fallback (usable when L2/L3 reject the Ds532 output), and the L4 explicit time budget (graceful degradation instead of timeout).

1.5b V0.3.3 morphology cleanup + bone fallback after bbox sanity fail

V0.3.3 patches the L2 + L3 path to harden the Ds532 argmax masks against a specific Ds532 failure mode where the dominant CC passes post-pad bbox sanity but is contaminated by sparse outlier CCs (thin streaks / isolated voxels) that survive the largest-CC keep heuristic via voxel-adjacent bridges. Two changes:

- **Ds532 mask morphology cleanup.** After argmax, each anatomy mask is passed through `binary_opening` (one iteration, 2x2x2 structure) and then a small-CC drop step (CCs with fewer than `MIN_DS532_CC_VOXELS = 500` voxels are removed). This is a benign cleanup on well-behaved cases (the dominant CC is unchanged) and a meaningful denoising on cases where Ds532 emits sparse outlier streaks.
- **Bone-skeleton fallback after post-pad bbox-sanity fail.** If the L3 pad-shrink cascade (24 -> 12 -> 6 -> 0 voxels) exhausts all four pad values and the post-pad bbox is still above the 50% threshold, V0.3.3 attempts the L1 bone-skeleton mask (largest CC only) before emitting zero for that anatomy. If the bone-skeleton bbox passes sanity, it is used as the Stage 2 ROI; otherwise the anatomy is emitted as zero (V0.3.1 behavior).

A/B (V0.3.3 vs V0.3.1) on a 3-case small-FOV pelvic slice (physical-x 248-258 mm; cases 162, 163, 189) shows **zero cohort delta on all metrics** (Fracture Dice, Instance F1, HD95 all 0.000). The morphology cleanup fires on case 162 (5 sparse CCs dropped on Sacrum, 5 on RightHip, 1 on LeftHip; 600-1000 voxels removed per anatomy) but the dominant CC is unchanged, so the painted ROI and downstream metrics are byte-identical to V0.3.1. The bone-skeleton fallback path does not fire on this slice because the Ds532 sanity check already passes at the L3 stage. V0.3.3 is therefore a **safety-net change**: benign on the tested cohort (no regression), but provides defense-in-depth against a known Ds532 sparse-outlier failure mode that does not appear in the held-out probe set.

2. Training

Item	Value
Stage 1 dataset	`Dataset532_PelvicAnatomyV2` (background / Sacrum / LeftHipbone / RightHipbone)
Stage 2 dataset	`Dataset537_PelvicBICMFragmentV5` (Sacrum / LeftHipbone / RightHipbone; no Femur)
Stage 2 trainer	`PengwinTrainerBoundaryFragmentSidecarCoreRecallDenseCandidateCore025StrongPeakNoContactABBCV291`
Plans	`nnUNetResEncUNetLPlans`
Config	`3d_fullres`
Stage 2 input channels	3 (bone-LUT CT, Ds532 anatomy prob, signed distance)
Fold	`fold_0` only
Boundary class weight (bw)	10
Epochs	50
Iterations / epoch	100 train / 20 val
Train ROIs	12 stratified (sub-fulltrain probe)

Item	Value
Val ROIs	6 stratified
Best EMA val score	0.188 (best checkpoint at epoch 34 spike)
Stage 2 checkpoint	`checkpoint_best.pth` (~819 MB)

The Stage 2 model was trained as a **probe** for the V291 ABBC head design, not as a full-data training run. A Phase 6 fulltrain on all 174 cases is planned but is not part of this submission.

3. Inference

3.1 Pipeline

1. **Read** input MHA from `/input/images/peripelvic-fracture-ct/`.
2. **LPS canonicalization** -- reorient the input volume into LPS and record the original orientation transform.
3. **Bone HU clip + LUT normalize** -- clip CT to `[-1000, 2000]` and apply the bone-window LUT (used by both stages).
4. **Stage 1 (Ds532 anatomy classifier)** -- nnU-Net ResEnc-L predict on the full CT, producing a 4-class softmax (background / Sacrum / LeftHipbone / RightHipbone).
5. **Per-anatomy ROI loop** for each of {Sacrum, LeftHipbone, RightHipbone}: a. Compute the bounding box of `prob >= 0.5` for this anatomy and pad by 24 voxels on each axis. b. Build a 3-channel ROI input:

- channel 0: bone-LUT CT crop,
- channel 1: Stage 1 anatomy probability crop,
- channel 2: signed distance to the `prob >= 0.5` surface in mm, clipped to +/- 40 mm. c. **Stage 2 (V291 ABBC) predict** -- nnU-Net ResEnc-L on the 3-channel ROI, producing a 4-class ABBC softmax (background / border / boundary / core). d. **Core-seed watershed** (V288-style decoder) lifts the ABBC softmax to per-instance integer fragments. e. **Connected-component prune** at ~1 cm^3 removes speckle. f. **Anatomy-size-ratio merge** with `sr_keep = 0.05` (fragments below 5% of the largest fragment within an anatomy are merged into the nearest large fragment). g. **Re-label** the fragments into the PENGWIN range for this anatomy: Sacrum 1-50, LeftHipbone 51-100, RightHipbone 101-150. h. Paste the relabeled ROI into the full-volume LPS label map.

6. **Femur (151-200)** is not modeled in V0.2 and remains 0.
7. **Reorient** the label volume from LPS back to the original input frame.
8. **Write** the integer 3D label map MHA to `/output/images/peripelvic-fracture-ct-segmentation/`, preserving the input spacing, dimensions and orientation.

The runtime entrypoint is `inference/inference.py`, packaged at `/opt/app/` in the container. Stage 1 (Ds532) and Stage 2 (V291) checkpoints, `plans.json`, and `dataset_fingerprint.json` are shipped via the model tarball, extracted by Grand Challenge under `/opt/ml/model/`.

4. Known limitations

- **Femur not modeled.** Femur fragments (label range 151-200) are written as background in V0.2. Any case whose ground-truth contains labels in [151, 200] will score zero on those instances. Femur is planned for V1.
- **Sub-fulltrain probe.** Only 12 of 174 cases were used to train the Stage 2 model; the EMA val score (0.188) and the hard-trio metrics (below) are well below the leaderboard target. Hard-trio Fracture Dice 0.812 is below the ~0.90 leaderboard expectation.
- **Single fold, single seed.** No ensembling, no TTA beyond nnU-Net defaults.
- **Phase 6 fulltrain planned.** The next submission will train on all 174 cases with both pelvic and femur anatomies and a full 1000-epoch schedule.

5. V0.2 held-out cohort metrics

Metrics are computed by our internal CLI `task1-v288-eval-v2` (`compute_task1_official_aligned_v2_metrics`), which is aligned to the PENGWIN 2026 official scoring spec.

The honest held-out cohort is the **five cases that are out-of-fold0 for the V0.2 Stage 2 model**: `001`, `002`, `004`, `005`, `006`. These are the only cases in this section that the model has never seen during training; they are the relevant signal for what the preliminary leaderboard will look like. Case `003` was inside the fold0 train set and is shown alongside as a sanity check, not as a leaderboard estimate.

Cohort	n	Fracture Dice	Local Dice (20mm)	HD95 (mm)	ASSD (mm)	Instance F1	Instance Recall	Instance Precision
Held-out (out-of-fold0)	**5** (001/002/004/005/006)	**0.612**	**0.616**	**26.7**	**5.7**	**0.683**	**0.785**	**0.708**
Train leak (in fold0 train)	1 (003)	0.827	0.828	3.8	1.5	0.808	0.952	0.778

The held-out row is the configuration actually shipped in the V0.2 container (**bw=10, sr_keep=0.05**, V291 ABBC Stage 2 + Ds532 Stage 1). The `003` train-leak row is shown only for comparison and must not be read as a held-out estimate.

GC preliminary score expectation: Fracture Dice ~0.55-0.65, Instance F1 ~0.60-0.70 (femur cases 151-200 will dilute the cohort because V0.2 emits zeros for that range).

Footnote -- *training-set fit only, not held-out*: an earlier Phase 5.5 v2 hard-trio (cases `003`, `011`, `017`, all in fold0 train per `splits_final.json`) reported Fracture Dice **0.812**, Local Dice **0.819**, Instance F1 **0.890**, HD95 **11.79 mm**, ASSD **2.59 mm**, Merge Err **2**, Split Err **21**, Topology Cons. **0.444** for the **bw 10 / sr 0.05 (V0.2)** configuration. These three cases are inside the fold0 training set and therefore are **not a held-out estimate**; they are retained here only to document the configuration sweep and should be ignored when projecting leaderboard performance. Alternate ablation configurations (bw 2.5 / sr 0.00, bw 2.5 / sr 0.05, bw 10 / sr 0.00) were run internally but are not reported here; only bw=10 / sr_keep=0.05 is shipped in the container.

5b. V0.3.1 held-out cohort metrics (this submission)

This is the **headline** held-out result for the V0.3.1 container that is actually shipped. The held-out cohort is identical to the V0.2 cohort -- the **five out-of-fold0 cases** `001`, `002`, `004`, `005`, `006` -- so the V0.3.1 row below is a like-for-like replacement of the V0.2 held-out row in Section 5; the V0.2 table is retained above for diff/context only.

Cohort	n	Fracture Dice	Local Dice (20mm)	HD95 (mm)	ASSD (mm)	Instance F1	Instance Recall	Instance Precision
V0.3.1 held-out (out-of-fold0)	**5** (001/002/004/005/006)	**0.621**	**0.616**	**20.6**	**4.96**	**0.805**	**0.774**	**0.933**
V0.2 held-out (out-of-fold0)	5 (001/002/004/005/006)	0.612	0.616	26.7	5.72	0.683	0.785	0.708
Delta (V0.3.1 - V0.2)	--	**+0.010**	0.000	**-6.1**	**-0.76**	**+0.123**	-0.011	**+0.225**
Train leak (in fold0 train, 003)	1 (003)	0.827	--	3.8	--	0.974	--	--

The V0.3.1 row is what we expect the preliminary leaderboard to look like for this submission. The Train-leak row (case `003`) is shown only as a sanity check that the model has not regressed on training data; it is **inside the fold0 train set** and must not be read as a held-out estimate.

Headline movements vs V0.2:

- **Fracture Dice 0.612 -> 0.621** (+0.010): small overall-overlap improvement.
- **Local Dice (20 mm) 0.616 -> 0.616** (0.000): unchanged, as expected -- L1-L4 robustness layers do not change the local 20 mm overlap structure on cases where V0.2 already produced a reasonable ROI.
- **HD95 26.7 mm -> 20.6 mm** (-6.1 mm) and **ASSD 5.72 mm -> 4.96 mm** (-0.76 mm): boundary-distance metrics improve substantially, consistent with L3 bbox-fraction sanity removing spurious far-field fragments.
- **Instance F1 0.683 -> 0.805** (+0.123) driven almost entirely by **Instance Precision 0.708 -> 0.933** (+0.225) while **Instance Recall 0.785 -> 0.774** (-0.011) stays roughly flat. This is the signature of the L2 argmax + L3 bbox-fraction stack: false positives from Ds532 OOD blow-ups are eliminated, recall is essentially preserved.

Per-case (V0.3.1 held-out):

Case	Fracture Dice	Instance F1	HD95 (mm)	Notes
001	0.620	0.667	15.4	--
002	0.688	0.889	8.0	--
004	0.483	0.822	29.5	hardest case
005	0.750	1.000	38.6	--
006	0.566	0.650	11.7	--

Case `004` is the hardest of the held-out cohort (lowest Fracture Dice at 0.483, highest HD95 at 29.5 mm), although it still attains a respectable Instance F1 of 0.822. Case `005` attains perfect Instance F1 (1.000) but with a high HD95 (38.6 mm), indicating that fragment identities are correctly recovered while the boundary location of one or more fragments is locally off.

Train-leak (case `003`, **training-set fit only, not held-out**): Fracture Dice **0.827**, Instance F1 **0.974**, HD95 **3.8 mm**. This case was inside the fold0 training set and is reported only as a sanity check that the V0.3.1 robustness layers (L1-L4) do not regress training-set fit. It must **not** be read as a held-out estimate or as a leaderboard projection.

V0.3.1 is the configuration actually shipped in this submission's container: V0.2 two-stage pipeline (Ds532 Stage 1 + V291 ABBC Stage 2, **bw=10, sr_keep=0.05**) wrapped in the L1-L4 robustness stack described in Section 1.5. The V0.3.1 numbers above supersede the V0.2 numbers in Section 5 as the headline held-out result for this submission; the V0.2 table is retained above only to make the diff explicit.

6. Pointers

- Stage 1 anatomy dataset: `Dataset532_PelvicAnatomyV2` (background / Sacrum / LeftHipbone / RightHipbone).
- Reference baseline: `YzzLiu/PENGWIN2026_Task1_AutoSeg_Baseline` (preprocessing / training / inference scaffolding; no `evaluate.py`).
- Internal eval CLI used to produce the metrics in Section 5: `task1-v288-eval-v2` (`compute_task1_official_aligned_v2_metrics`).
- Internal predict CLI mirrored by the container entrypoint: `task1-v288-predict` (`code_task1/eval.py`, `--cases / --out-root / --checkpoint / --fold / --trainer / --plans / --config / --preprocessed-plans-dir`).